RDBMS Study Material

Class 12 IT/ITES Vocational Course
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> **Teacher's Note:** Welcome, students! Mastering Relational Database Management Systems (RDBMS) using SQL is a core skill for IT professionals. This guide covers essential Data Query Language (DQL), Data Control Language (DCL), and Transaction Control Language (TCL) concepts required for your CBSE Class 12 board exams and practicals.

1. The Practice Table: IT_STUDENTS

For all the examples in this module, we will be using the **IT_STUDENTS** table. It contains information about students enrolled in various vocational streams, their marks, locations, and scholarship status.

RollNo	Name	Stream	Marks	City	Scholarship
101	Aarav	IT	85	Delhi	5000
102	Priya	ITES	92	Mumbai	NULL
103	Rohan	IT	78	Delhi	2000
104	Sneha	ITES	88	Chennai	NULL
105	Kabir	IT	95	Mumbai	8000
106	Ananya	ITES	72	Delhi	1500

2. Data Retrieval Clauses and Operators (DQL)

The AS Clause (Aliasing)

The AS keyword is used to temporarily rename a column or table in the output for better readability.

```
SELECT Name AS Student_Name, Marks AS Total_Marks FROM IT_STUDENTS;
```

The DISTINCT Clause

The `DISTINCT` keyword is used to return only distinct (different) values. It eliminates duplicate rows from the results.

```
SELECT DISTINCT City FROM IT_STUDENTS;
```

> **Output:** Delhi, Mumbai, Chennai

Logical Operators: AND, OR, NOT

These operators are used to combine multiple conditions in a `WHERE` clause.

```
-- AND: Both conditions must be true
SELECT Name FROM IT_STUDENTS WHERE Stream = 'IT' AND City = 'Delhi';

-- OR: At least one condition must be true
SELECT Name FROM IT_STUDENTS WHERE City = 'Chennai' OR City = 'Mumbai';

-- NOT: Reverses the condition
SELECT Name FROM IT_STUDENTS WHERE NOT City = 'Delhi';
```

The BETWEEN Operator

The `BETWEEN` operator selects values within a given range (inclusive of endpoints).

```
SELECT Name, Marks FROM IT_STUDENTS WHERE Marks BETWEEN 80 AND 90;
```

> **Output:** Aarav (85), Sneha (88)

The IN Operator

The `IN` operator allows you to specify multiple possible values for a column (like a shorthand for multiple ORs).

```
SELECT Name, City FROM IT_STUDENTS WHERE City IN ('Mumbai', 'Chennai');
```

String and Substring Searches (LIKE Operator)

The `LIKE` operator is used for pattern matching.
`%` represents zero, one, or multiple characters.
`_` (underscore) represents exactly one character.

```
-- Starts with 'A'
SELECT Name FROM IT_STUDENTS WHERE Name LIKE 'A%';

-- Ends with 'a'
SELECT Name FROM IT_STUDENTS WHERE Name LIKE '%a';

-- Has 'o' as the second letter
SELECT Name FROM IT_STUDENTS WHERE Name LIKE '_o%';
```

IS NULL and IS NOT NULL

In SQL, a `NULL` value means "no value" or "missing data". You cannot use `=` or `!=` to check for `NULL`. Instead, you must use `IS NULL` or `IS NOT NULL`.

```
-- Find students with no scholarship
SELECT Name FROM IT_STUDENTS WHERE Scholarship IS NULL;

-- Find students who have received a scholarship
SELECT Name, Scholarship FROM IT_STUDENTS WHERE Scholarship IS NOT NULL;
```

The ORDER BY Clause

The `ORDER BY` keyword is used to sort the result-set in ascending (`ASC`) or descending (`DESC`) order.

```
SELECT Name, Marks FROM IT_STUDENTS ORDER BY Marks DESC;
```

3. Grouping Data

The GROUP BY Clause

The `GROUP BY` statement groups rows that have the same values into summary rows, typically to be used with aggregate functions like `COUNT()`, `MAX()`, `MIN()`, `SUM()`, `AVG()`.

```
SELECT Stream, COUNT(*) AS Total_Students, AVG(Marks) AS Average_Marks
FROM IT_STUDENTS
GROUP BY Stream;
```

The HAVING Clause

The `HAVING` clause was added to SQL because the `WHERE` keyword cannot be used with aggregate functions. It filters records after the grouping has occurred.

```
SELECT City, COUNT(*) AS Student_Count
FROM IT_STUDENTS
GROUP BY City
HAVING COUNT(*) > 1;
```

4. DCL (Data Control Language) Commands

DCL commands are used to manage permissions and access control to the database.

- **GRANT:** Gives user access privileges to the database.
- **REVOKE:** Takes back permissions granted to the user.

```
-- Granting read and insert permissions to 'student_user'
GRANT SELECT, INSERT ON IT_STUDENTS TO student_user;

-- Revoking the insert permission
REVOKE INSERT ON IT_STUDENTS FROM student_user;
```

5. TCL (Transaction Control Language) Commands

TCL commands are used to manage transactions in the database. A transaction is a logical unit of work (e.g., updating a student's marks and scholarship together). It ensures the **ACID** (Atomicity, Consistency, Isolation, Durability) properties.

- **COMMIT:** Permanently saves all the changes made during the current transaction.
- **ROLLBACK:** Undoes the changes made in the current transaction (restores the database to its previous state).
- **SAVEPOINT:** Sets a point within a transaction to which you can later roll back, without rolling back the entire transaction.

```
BEGIN;
UPDATE IT_STUDENTS SET Marks = Marks + 2 WHERE City = 'Delhi';
SAVEPOINT mark_updated;

DELETE FROM IT_STUDENTS WHERE RollNo = 106;
-- Wait, that deletion was a mistake! Let's roll back to the savepoint.
ROLLBACK TO mark_updated;

-- Save the marks update permanently.
COMMIT;
```